ASSIGNMENT 25

gitlab ci- apply and destroy

introduction :

The .gitlab-ci.yml file is a YAML-based configuration file that defines the structure and execution of the Continuous Integration/Continuous Delivery (CI/CD) pipelines in GitLab. It is placed in the root directory of a GitLab repository and tells the GitLab Runner application what tasks to perform, and in what order.

Key Components

The file uses specific keywords to define the CI/CD workflow:

Jobs: These are the fundamental units of work, defining specific tasks such as compiling code, running tests, or deploying an application. Each job contains a script section with the commands to be executed.

Stages: Stages define the order in which jobs run. Jobs within the same stage run in parallel, and if all succeed, the pipeline proceeds to the next stage. A typical pipeline might have build, test, and deploy stages.

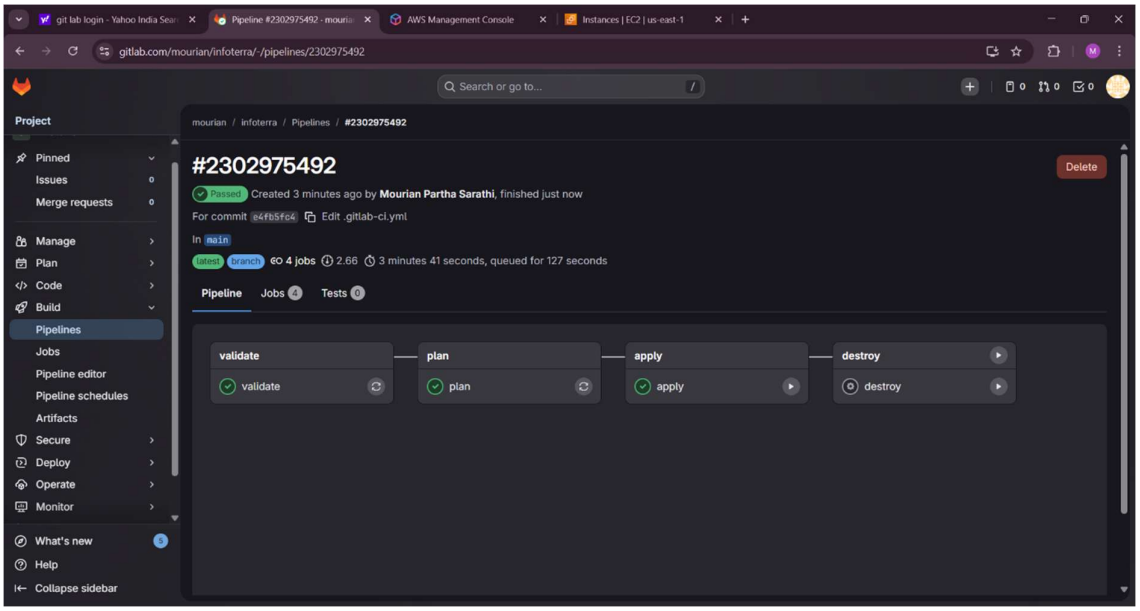
Runners: These are the agents that actually execute the jobs defined in the .gitlab-ci.yml file on computing infrastructure.

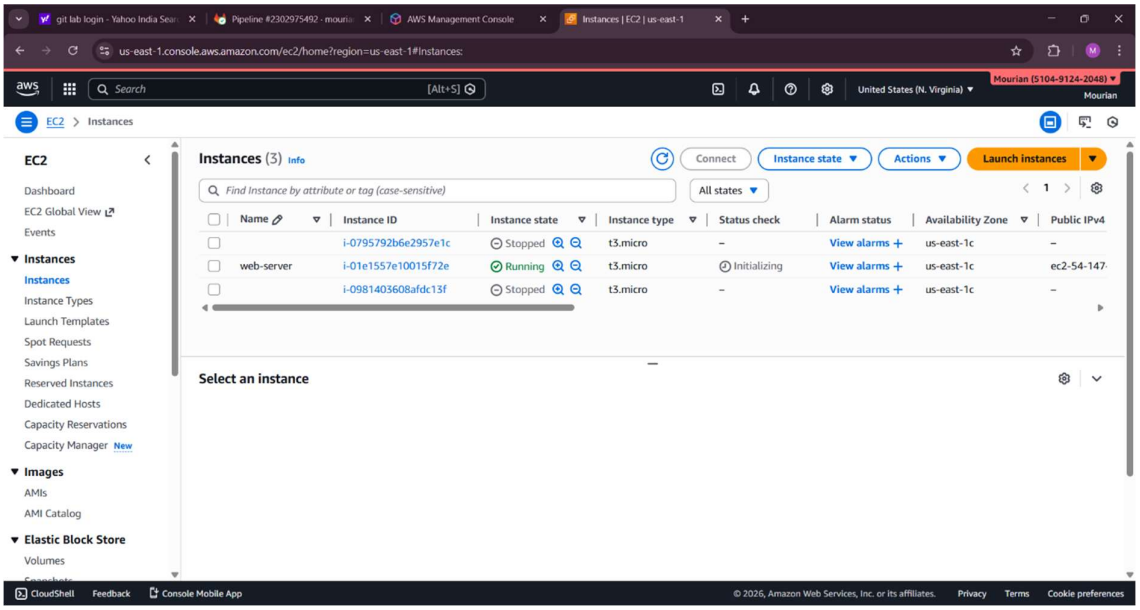
Variables: You can define custom CI/CD variables or use predefined variables to store values (e.g., environment URLs, API keys) and control pipeline behavior dynamically.

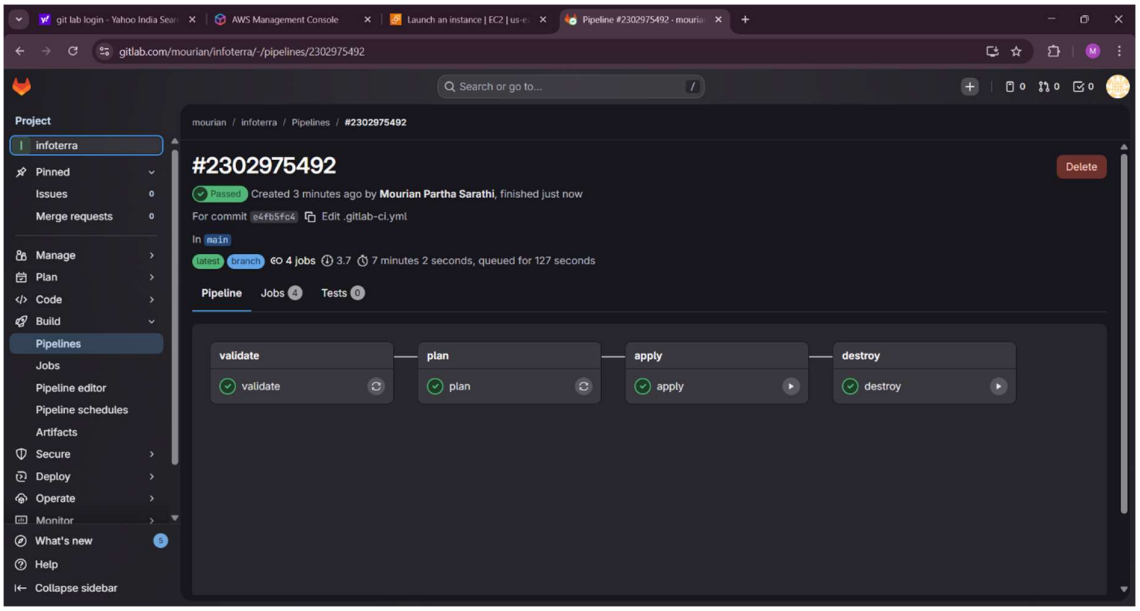
Artifacts and Cache: The artifacts keyword specifies files that should be saved and passed between jobs and stages, while cache helps speed up subsequent runs by storing dependencies.

How it works

When a developer pushes code to the repository (or another trigger event occurs, such as a merge request), GitLab automatically detects the .gitlab-ci.yml file and initiates a pipeline. The GitLab Runner reads the file and executes the jobs according to the defined configuration. The status and logs for each job can be viewed in the "CI/CD > Pipelines" section of the GitLab project interface.


gitlab.com/mourian/infoterra/-/pipelines/2302975492
Project
mourian / infoterra / Pipelines / #2302975492
#2302975492
Delete
Passed Created 3 minutes ago by Mourian Partha Sarathi, finished just now
For commit e4fb5fc4 Edit .gitlab-ci.yml
In main
latest branch 4 jobs 2.66 3 minutes 41 seconds, queued for 127 seconds
Pipeline
Jobs 4
Tests 0
validate
validate
plan
plan
apply
apply
destroy
destroy
Pinned
Issues
Merge requests
Manage
Plan
Code
Build
Pipelines
Jobs
Pipeline editor
Pipeline schedules
Artifacts
Secure
Deploy
Operate
Monitor
What's new
Help
Collapse sidebar


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
United States (N. Virginia)
Mourian (5104-9124-2048)
Mourian
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instances (3) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
i-0795792b6e2957e1c Stopped t3.micro – View alarms + us-east-1c –
web-server i-01e1557e10015f72e Running t3.micro Initializing View alarms + us-east-1c ec2-54-147-
i-0981403608afdc13f Stopped t3.micro – View alarms + us-east-1c –
Select an instance
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences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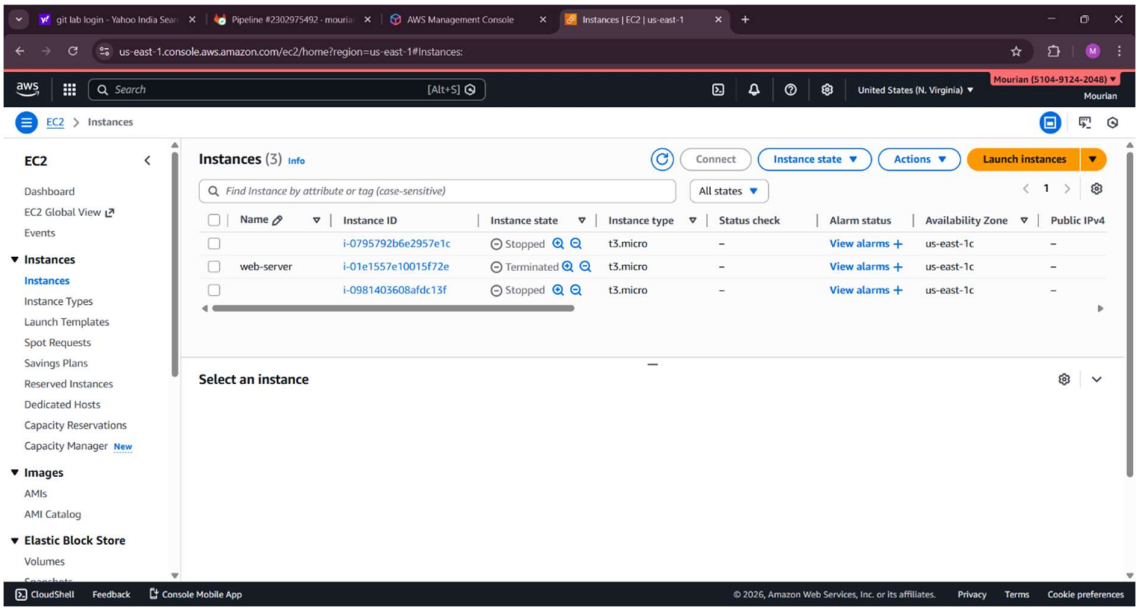
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Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instances (3) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
i-0795792b6e2957e1c Stopped t3.micro – View alarms + us-east-1c –
web-server i-01e1557e10015f72e Terminated t3.micro – View alarms + us-east-1c –
i-0981403608afdc13f Stopped t3.micro – View alarms + us-east-1c –
Select an instance
CloudShell
Feedback
Console Mobile App
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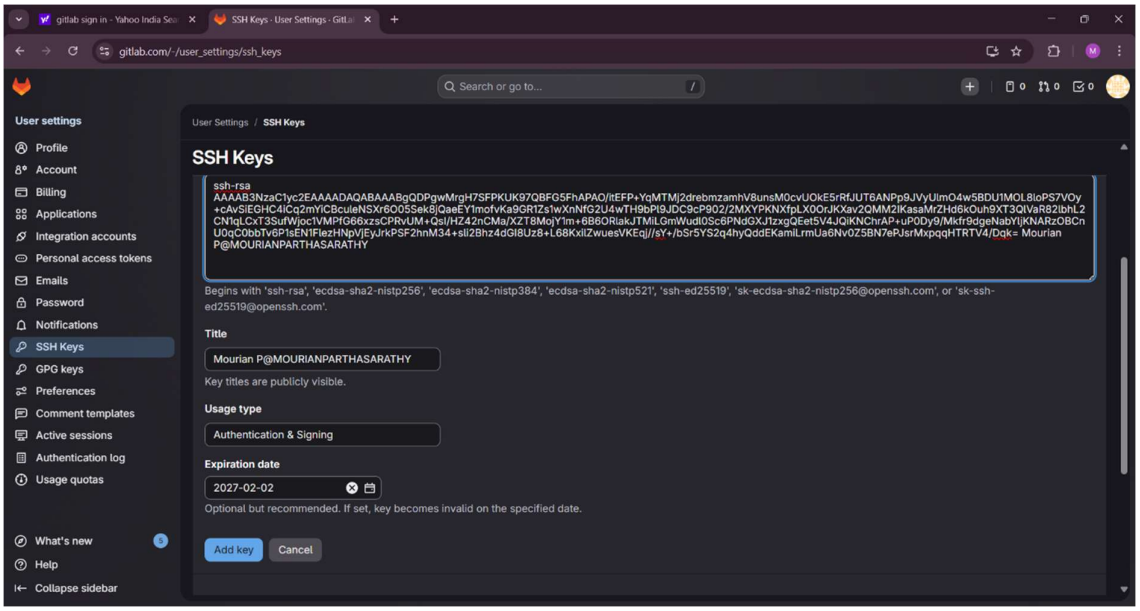
Privacy
Terms
Cookie preferences

First we create ssh keys to set our git lab account then oly we work to connect instance .



This is the code to destroy our instance automatically in our gitlab :

```
stages:
  - validate
  - plan
  - apply
  - destroy

image: alpine:3.18

before_script:
```

```
  - apk add --no-cache bash curl unzip
  - curl -LO https://releases.hashicorp.com/terraform/1.5.0/terraform_1.5.0_linux_amd64.zip
  - unzip terraform_1.5.0_linux_amd64.zip
  - mv terraform /usr/local/bin/
  - terraform --version

validate:
  stage: validate
  script:
    - cd newtech
    - terraform init
    - terraform validate

plan:
  stage: plan
  script:
    - cd newtech
    - terraform init
    - terraform plan -out=tfplan
  artifacts:
    paths:
      - newtech/tfplan

apply:
  stage: apply
  when: manual
  script:
    - cd newtech
    - terraform init
    - terraform apply -auto-approve tfplan
  artifacts:
```

```
    paths:
      - newtech/terraform.tfstate
      - newtech/terraform.tfstate.backup
destroy:
  stage: destroy
  when: manual
  dependencies:
    - apply
  script:
    - cd newtech
    - terraform init
    - terraform destroy -auto-approve
```